

2) $X_i \sim \text{Gamma}(k, \lambda)$

$i = 1, \dots, n$

p.d.f of X_i is given by

$$f(x; k, \lambda) = \begin{cases} \frac{1}{\Gamma(k)} x^{k-1} \lambda^k \exp(-\lambda x), & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Its moment generating function (M_X) is given by

$$M_{X_i}(t) = (1 - \lambda t)^{-k}$$

$$t < \frac{1}{\lambda}$$

~~$M_{X_i}(t)$~~

$$\text{Let } Y = \sum_{i=1}^n X_i$$

then

$$M_Y(t) = \prod_{i=1}^n (1 - \lambda t)^{-k}$$

$$= (1 - \lambda t)^{-nk}$$

So,

$$Y \sim \text{Gamma}(nk, \lambda)$$

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$$\sum_{i=1}^n X_i \sim \text{Gamma}(nk, \lambda)$$

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$$E\left[\frac{\sum_{i=1}^n X_i}{n}\right] = \frac{nk\lambda}{n} = k\lambda$$

$$\begin{aligned}
 V\left(\frac{\sum_{i=1}^n X_i}{n}\right) &= E\left[\left(\frac{Y}{n} - E\bar{X}_n\right)\left(\frac{Y}{n} - E\bar{X}_n\right)\right] \\
 &= E\left[\left(\frac{Y}{n} - \frac{EY}{n}\right)\left(\frac{Y}{n} - \frac{EY}{n}\right)\right] \\
 &= \frac{V(Y)}{n^2} = \frac{\alpha \lambda^2}{n^2}
 \end{aligned}$$

$$AB = \alpha \lambda^2$$

$$AB^2 = \frac{\alpha \lambda^2}{n^2}$$

$$B = \frac{\lambda}{n^2}$$

$$A = n^2 \alpha$$

So,

~~$$\bar{X}_n \sim \text{Gamma}(n^2 \alpha, \frac{\lambda}{n^2})$$~~

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$$\begin{aligned}
 3) \quad b_x(x) &= 2 \left(\frac{1}{2} \int_0^\infty \frac{1}{y} e^{(-y - \frac{x}{y})} dy \right) \\
 &= \int_0^\infty \frac{1}{y} e^{(-y - \frac{x}{y})} dy \\
 \int_0^\infty b_x(x) dx &= \int_0^\infty \int_0^\infty \frac{1}{y} e^{(-x - \frac{x}{y})} dy dx \\
 &= \int_0^\infty \frac{1}{y} e^{-x} \int_0^\infty e^{-\frac{x}{y}} dx dy \\
 &= \int_0^\infty \frac{1}{y} e^{-x} \left(\frac{1}{y} (0 - (-1)) \right) dy \\
 &= \int_0^\infty e^{-x} dy = 1
 \end{aligned}$$

$$X|Y \sim F, Y \sim G$$

$$b_{x|y}(x|y) = \frac{b_{x,y}(x,y)}{b_y(y)}$$

$$\rightarrow f(x,y) = f(x|y) f(y)$$

$$\begin{aligned}
 \int_0^\infty \int_0^\infty \frac{1}{y} e^{-x - \frac{x}{y}} dy dx &= f(x,y) \\
 \int_0^\infty \frac{1}{y} e^{-y} dy &= f(y) \\
 \int_0^\infty e^{-x/y} dx &= f(x|y)
 \end{aligned}$$

$$b(x) = \int_0^{\infty} \frac{1}{y} \exp(-y - x/y) dy.$$

$$b(x, y) = \underbrace{\int_0^{\infty} \exp(-y)}_{\rightarrow \exp(1)} \cdot \underbrace{\frac{1}{y} \exp(-x/y)}_{\rightarrow \exp(y)}$$

6)

x

$$m(y, \dots, x) =$$

$$\int_0^{\infty} \frac{e^{-x} x^y}{y!} dx$$

$$= \left[\frac{x^y}{y!} - \frac{1}{y!} x^{y-1} \right]_0^{\infty}$$

$$= \frac{\Gamma(y+1)}{y!} = y+1$$

$$f(x, y) = \frac{\frac{1}{\Gamma(a)} x^{a-1} b^{-a} \exp\left(-\frac{x}{b}\right) \exp(-x) x^y}{\Gamma(y)}$$

$$y+1$$

$$= \frac{x^{a+y-1} b^{-a} \exp\left(-x\left(\frac{1+b}{b}\right)\right)}{\Gamma(a) \Gamma(y+1)}$$